

Science not only changes what we know, but also how we think about our place in the cosmos. I invite you to take a moment to discover our world at night through the eyes of space science. I have no doubt that inherent beauty of these images will inspire—feeding the soul and mind of the reader. As Vincent van Gogh said, “For my part I know nothing with any certainty, but the sight of the stars makes me dream.” I challenge you to look back from the stars to see Earth in a completely new way.

Thomas H. Zurbuchen

Associate Administrator

NASA Science Mission Directorate

Preface

To keen observers, the nocturnal Earth is not pitch black, featureless, or static. The stars and the Moon provide illumination that differs from, and complements, daylight. Natural Earth processes such as volcanic eruptions, auroras, lightning, and meteors entering the atmosphere generate localized visible light on timescales ranging from subsecond (lightning), to days, weeks (forest fires), and months (volcanic eruptions).

Most interesting and unique (as far as we know) to Earth, is the nighttime visible illumination emitted from our planet that is associated with human activities. Whether purposefully designed to banish darkness (such as lighting for safety, industrial activities, commerce, and transportation) or a secondary result of (such as gas flares associated with mining and hydrocarbon extraction activities, or nocturnal commercial fishing), anthropogenic sources of nighttime light are often broadly distributed in space and sustained in time—over years and even decades. Because these light sources are inextricably tied to human activities and societies, extensive and long-term measurement and monitoring of Earth’s anthropogenic nocturnal lights can provide valuable insights into the spatial distribution of our species and the ways in which society is changing—and is changed by—the environment on a wide range of time scales.

Over the past four decades, sensitive imaging instruments have been operated on low-Earth-orbiting satellites to measure natural and human-caused visible nocturnal illumination, both reflected and Earth-generated. The satellite sensors provide unique imagery: global coverage yet with high spatial resolution, and frequent measurements over long periods of time.

The combined, multisatellite global nocturnal illumination dataset contains a treasure trove of unique information about our planet and our species—and the

interactions between society and natural processes. Beyond academic study, Earth nightlight measurements are being used to help save lives and property around the globe, by allowing accurate identification and monitoring of ongoing events like eruptions and fires even in remote locations, and by pinpointing and enabling quantitative tracking of regions of power outage and recovery following extreme weather events and geohazards in populated areas.

Earth at Night tells—in the words of the women and men, the scientists and engineers who are actually designing the instruments and conducting the analyses of Earth’s nocturnal illumination imagery—the story of satellite measurements of global light in the night. It shows how ever-increasing instrument capability has improved the sensitivity, accuracy, coverage, and resolution of the observations. Through striking illustrations and clear explanations, the book summarizes many examples of analyses from the satellite nightlight data record—examples that themselves shine light on the ever-changing environment and our human impact on Earth.

This artful volume reaffirms our human ability to harness technology and science to observe and understand Earth for the benefit of all humankind. Above all, it once again illuminates the beauty and majesty of our home planet—at all hours.

Michael H. Freilich

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An Introduction to Nightlights



Dazzling photographs and images from space of our planet's nightlights have captivated public attention for decades. In such images, patterns are immediately seen based on the presence or absence of light: a distinct coastline, bodies of water recognizable by their dark silhouettes, and the faint tendrils of roads and highways emanating from the brilliant blobs of light that are our modern, well-lit cities.



Blue Marble Becomes Black Marble

[Right] Dubbed the “Blue Marble,” this classic photograph of Earth was taken by the Apollo 17 crew on December 7, 1972.



The color of an object is actually the color of the light reflected while all other colors are absorbed. Most of the light that is reflected by clear, open ocean water is blue, while the red portion of sunlight is quickly absorbed near the surface. Therefore, very deep water with no reflections off the sea floor appears dark navy blue. Since approximately 70% of Earth's surface is covered by deep ocean water, Earth appears as a blue marble when viewed from space during daytime hours. Clouds suspended in Earth's atmosphere provide the white swirls.